

# Jiale Han

MACHINE LEARNING · STATISTICAL LEARNING · ADS & RECOMMENDATION · LLMs

Los Angeles, CA

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## EDUCATION

### University of California, Los Angeles (UCLA)

PH.D. IN STATISTICS

• **Advisor:** Prof. Xiaowu Dai    **GPA:** 3.98/4.00

• **Selected awards:** Most Promising Statistician Award (2 awarded department-wide per year), Summer Mentored Research Fellowship

Los Angeles, CA, U.S.

Sep. 2022 - Mar. 2027

### University of Chinese Academy of Sciences (UCAS)

B.S. IN MATHEMATICS AND APPLIED MATHEMATICS

• **GPA:** 3.87/4.00

• **Selected awards:** Overseas Graduate Studies Fellowship (Top 5%), Tang-Lixin Outstanding Student Leader Scholarship (one per class per year)

Beijing, China

Sep. 2018 - Jun. 2022

## PROFESSIONAL EXPERIENCES

### Roblox

APPLIED SCIENTIST INTERN - ADS DELIVERY TEAM

- **Developed a Thompson-sampling-based dynamic creative optimization framework** for ads retrieval, balancing exploration-exploitation while mitigating A/B-test data leakage; improved predicted play-through rate (PTR) by **8.14%** offline with **0.2%** added latency.
- **Launched to Roblox's production** sponsored ads system after an **online A/B test**: PTR **+1.4%**, CTR **+1.3%**, CPP **-0.3%**, ads value **+1.4%**.
- Summarized the work in a **first-author paper** under review at WWW 2027.

San Mateo, CA, U.S.

Jun. 2026 - Sep. 2026

### TikTok

SOFTWARE ENGINEER INTERN - LIVE DATA BP TEAM

- Contributed to an **LLM-powered data agent** by integrating enterprise BI knowledge bases with semantic retrieval and structured datasets.
- Benchmarked the agent with automated evaluation and tuning tools on curated question sets, and used LLMs to analyze results and iteratively **post-train** it.

San Jose, CA, U.S.

Sep. 2025 - Dec. 2025

### JD.com

ML RESEARCH INTERN - ADS MECHANISM TEAM

- Built an ML solution for **uncertainty quantification** of future ad-opportunity values, with a calibration method that improves bidding pacing when true values such as **CVR** are unknown.
- Derived theoretical guarantees for **auto-bidding** under non-i.i.d. requests and improved **CVR** by **4%** on JD.com data; first-author paper accepted to **WWW 2026 (oral)**.

Beijing, China

Jun. 2025 - Sep. 2025

## SELECTED RESEARCH

### Quality-Preserving LLM Advertising

- Designed a framework combining **RAG** and **auction theory** to trade off revenue against response quality in LLM-generated content.
- Submitted to The 41st AAAI Conference on Artificial Intelligence (AAAI), 2027.

Jan. 2026 - May 2026

### Variance Reduction via Resampling and Experience Replay

- Modeled experience replay through **resampling statistics** and proved rigorous **variance-reduction** guarantees for **RL** algorithms.
- Accepted to The 40th Annual AAAI Conference on Artificial Intelligence (AAAI), 2026 (Oral, top 4.5%).

Jun. 2024 - Jan. 2025

### Online Auction Design Using Uncertainty Quantification

- Designed an incentive-compatible **online auction** with a distribution-free seller-revenue guarantee for heterogeneous future bidders.
- Accepted to Journal of the American Statistical Association (JASA), 2025.

May 2023 - May 2024

## PUBLICATIONS

- **Jiale Han**, Xiaowu Dai. Online Auction Design Using Distribution-Free Uncertainty Quantification with Applications to E-Commerce. *Journal of the American Statistical Association*, 2025. 🔗 JASA 🔗 Code
- **Jiale Han**, Xiaowu Dai, Yuhua Zhu. Variance Reduction via Resampling and Experience Replay. *AAAI Conference on Artificial Intelligence*, 2026. (**Oral Presentation**, top 4.5%) 🔗 AAAI'2026 🔗 Code
- **Jiale Han**, Chun Gan, Chengcheng Zhang, Jie He, Zhangang Lin, Ching Law, Xiaowu Dai. Auto-Bidding under Return-on-Spend Constraints with Uncertainty Quantification. *The ACM Web Conference*, 2026. (**Oral Presentation**) 🔗 WWW'2026 🔗 Code
- **Jiale Han**, Xiaowu Dai. A Robust Multi-Item Auction Design with Statistical Learning. *ICLR Workshop on AI for Mechanism Design and Strategic Decision Making*, 2026. 🔗 ICLR AIMS'2026 🔗 Code
- Baiting Chen, Tong Zhu, **Jiale Han**, Lexin Li, Gang Li, Xiaowu Dai. Incentivizing Truthful Language Models via Peer Elicitation Games. *Advances in Neural Information Processing Systems*, 2025. 🔗 NeurIPS'2025 🔗 Code
- **Jiale Han**, Xiaowu Dai. Mechanism Design for Quality-Preserving LLM Advertising. *Submitted to AAAI*, 2027. 🔗 arXiv 🔗 Code

## SKILLS

### ML / Statistics

Uncertainty quantification, auto-bidding, bandits, A/B testing, reinforcement learning, mechanism design, LLMs

### Tools & Frameworks

TensorFlow, scikit-learn, XGBoost, Git, Business Intelligence tools, 飞书

### Programming

Python, R, and SQL